

Muhammad Umer

AI Engineer · Full-Stack Software Engineer

Date of birth: 28/08/2002 | **Nationality:** Pakistani | **Home:** Lahore (Pakistan)

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ABOUT ME

AI and full-stack software engineer with over two years of remote experience at US-based companies. I design and ship LLM agents (tool calling, RAG, Model Context Protocol), real-time voice AI and multi-tenant SaaS platforms end to end, from data model and API to CI/CD and production deployment on Docker and Kubernetes. BS Computer Science (CGPA 3.68/4.00) with final-year research in deep reinforcement learning for autonomous drone navigation.

WORK EXPERIENCE

01/2026 – CURRENT Remote (United States)

AI/ML Engineer & Full-Stack Developer Kindwell Solutions & Logicbuilder.ai (*sister companies, projects across both*)

- Sole engineer of **AlphaVenue.ai**, a multi-tenant SaaS for wedding and event venues: Express/TypeScript API (132 MongoDB data models), React web app (179 pages), Redis/BullMQ workers, and ~1,450 automated test files in sharded GitHub Actions CI/CD.
- Built **ELLA**, the platform's LLM assistant: OpenAI tool calling across 104 operational tools with human approval for write actions, retrieval-augmented generation over per-tenant knowledge bases (Qdrant), and an MCP server exposing role-scoped tools to external AI agents via OAuth.
- Developed a real-time voice agent (Pipecat: Deepgram STT → LLM → Cartesia TTS over Twilio) with live call monitoring, and replaced GoHighLevel with a native omnichannel CRM for voice, SMS and email.
- Primary engineer of **LogicOne Dialer** (logicone.ai), Logicbuilder's AI sales-calling platform (Next.js, Prisma/PostgreSQL, Twilio): grew it from 62 to 96 data models and 271 to 429 API routes, shipping predictive dialling, live call coaching, Pipecat/Gemini Live voice agents and Stripe Connect agency billing.
- Contributed nine merged pull requests to partner **Juno Innovations'** open-source Terra plugin catalogue for Kubernetes (Helm, Argo CD, Gateway API): new cert-issuer, external-dns and domain-manager plugins and custom-domain publishing.

03/2024 – 12/2025 Remote (United States)

Full-Stack Developer ArchiPartnerDesign

- Built **Elio** (elio.care), a construction marketplace for homeowners, contractors and vendors: Express/TypeScript API on MongoDB, Redis and Socket.io with 32 data models (projects, quotes, milestones, invoices, ledger, vendor payouts) and a React/TypeScript web app with Stripe and Square payments.
- Migrated the Flutter mobile app (admin, client, contractor and vendor roles) from Firebase to the platform's JWT-authenticated REST API and prepared the iOS and Google Play releases.
- Built the multilingual marketing site (React, Tailwind CSS; English, Spanish, Chinese) with automated deployment, and hardened the production VPS (nginx, TLS 1.2/1.3, HSTS).

08/2022 – CURRENT Lahore, Pakistan – remote clients

Freelance Software Engineer Self-employed

- Sole developer of a donation-management system for an educational academy (app.hria.edu.pk): Electron desktop app (Windows, macOS) on an Express/TypeScript/MongoDB API with 169 endpoints, thermal-receipt printing, TOTP 2FA, English/Urdu reports and CI-built auto-updating releases.
- Turned a prototype into a production AI quantity take-off pipeline for a construction estimating firm: Playwright drawing capture, Claude vision extraction routed per sheet type, and a Flask job dashboard; covered by 37 test modules and a golden-file regression harness.
- Co-developed a social-media scheduling SaaS (Next.js, Prisma, PostgreSQL): Facebook, Instagram, LinkedIn, X and YouTube publishing connectors, OAuth flows, BullMQ workers and Stripe billing.

11/2022 – 02/2023 Remote – C++ and Java (fundamentals, OOP) for university students in Perth, Australia

Programming Tutor (part-time, online) Private tutoring

PROJECTS

06/2026 – CURRENT restaurantos.softxlogic.com

RestaurantOS – multi-tenant restaurant ERP SaaS Technical lead, team of 4

- Architected and led a white-label ERP (point of sale, kitchen display, inventory, finance, purchasing, HR/payroll, reporting): 15 Java/Spring Boot microservices behind Spring Cloud Gateway, a Next.js front end, RabbitMQ with dead-letter queues, Redis and ClickHouse.
- Designed tenant isolation and security: PostgreSQL row-level security, fail-closed OPA/Rego authorisation with 100% policy-test coverage enforced in CI, RS256 JWT with TOTP 2FA; GitHub Actions delivery to Kubernetes (k3s).

EDUCATION AND TRAINING

02/2022 – 03/2026 Lahore, Pakistan
Bachelor of Science in Computer Science COMSATS University Islamabad
Level in EQF: EQF level 6 | Final grade: CGPA 3.68 / 4.00 | Credit hours: 133

- **Final-year thesis (team of 3, grade A):** *Autonomous Drone Navigation in Dense, GPS-Denied Environments.* Vision-based quadrotor navigation through dense forest in AirSim (Unreal Engine 5): a maximum-entropy Soft Actor-Critic policy acting on noisy depth images, trained against an asymmetric privileged critic that sees clean depth and state, and benchmarked against TD3 (PyTorch).
- **My contributions:** reward-shaping design (potential-based progress, obstacle proximity, heading and altitude terms), checkpointing and evaluation tooling, a gRPC pipeline streaming experience to a remote GPU trainer, and the ground-control desktop app (Electron, React/TypeScript, FastAPI) for live AirSim telemetry, mission control and model versioning.
- **Relevant modules:** Artificial Intelligence, Machine Learning, Computer Vision, Digital Image Processing, Robotics, Parallel and Distributed Computing, Linear Algebra, Multivariable Calculus, Probability and Statistics.

08/2018 – 09/2020 Toba Tek Singh, Pakistan
Higher Secondary School Certificate (Science) Punjab College of Science and Commerce
Level in EQF: EQF level 4 | Grades: Physics 96.5%, Chemistry 98.0%, Biology 96.5%

PUBLICATIONS

2026 Co-author · Manuscript completed, to be submitted for peer review · Extends the final-year thesis
Maximum-Entropy Privileged Reinforcement Learning for Vision-Based Drone Navigation in Dense Forests

- Reports 87% navigation success in a simulated 4,008-tree forest at nine times the obstacle-count density of the prior work it builds on; an ablation shows that withholding the privileged critic cuts success from 87% to 37%, and the SAC policy beats the TD3 baseline at every density tested.

DIGITAL SKILLS

AI & machine learning: PyTorch, TensorFlow/Keras, deep reinforcement learning (TD3, SAC), computer vision (OpenCV), LLM agents and tool calling, retrieval-augmented generation (RAG), vector databases (Qdrant), Model Context Protocol (MCP), OpenAI, Anthropic Claude and Gemini APIs, voice AI (Pipecat, Deepgram, Twilio)

Programming languages: Python, TypeScript, JavaScript, Java, C++, C#, Dart, SQL

Back end & data: Node.js/Express, Spring Boot/Spring Cloud, FastAPI, Flask, REST, WebSockets, gRPC, PostgreSQL, MongoDB, Redis, RabbitMQ, BullMQ, Prisma

Front end, mobile & desktop: React, Next.js, Tailwind CSS, Flutter, Electron

Cloud & DevOps: Docker, Kubernetes (k3s), Helm, GitHub Actions CI/CD, Linux server administration (nginx, PM2), OAuth 2.0/JWT, Open Policy Agent (OPA)

LANGUAGE SKILLS

Mother tongue(s): Urdu | English: IELTS Academic, overall band 7.0 (CEFR C1), 07/2026; other languages self-assessed

Other language(s)	Listening	Reading	Spoken production	Spoken interaction	Writing
English IELTS 7.0	B2	C2	B2	B2	B2
Punjabi	C2	C2	C2	C2	B1
German	A1	A1	A1	A1	A1

HONOURS, AWARDS AND CERTIFICATIONS

01/2024 Higher Education Commission, Government of Pakistan
Prime Minister’s Youth Laptop Scheme – national merit award Awarded on academic merit

02/2025 Google, via Coursera
Foundations of Data Science coursera.org/verify/74H22H9UABHK

VOLUNTEERING

04/2022 – 10/2025 Lahore, Pakistan
Co-founder – Helping Hands, student-led welfare initiative

- Organised fundraising, food-distribution and seasonal relief drives for low-income families.